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The sphere packing problem in high dimensions

A physicist's view

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The sphere packing problem

Motivations

THE SPHERE PACKING PROBLEM

N. J. A. SLOANE

<https://arxiv.org/pdf/math/0207256>

ABSTRACT. A brief report on recent work on the sphere-packing problem.

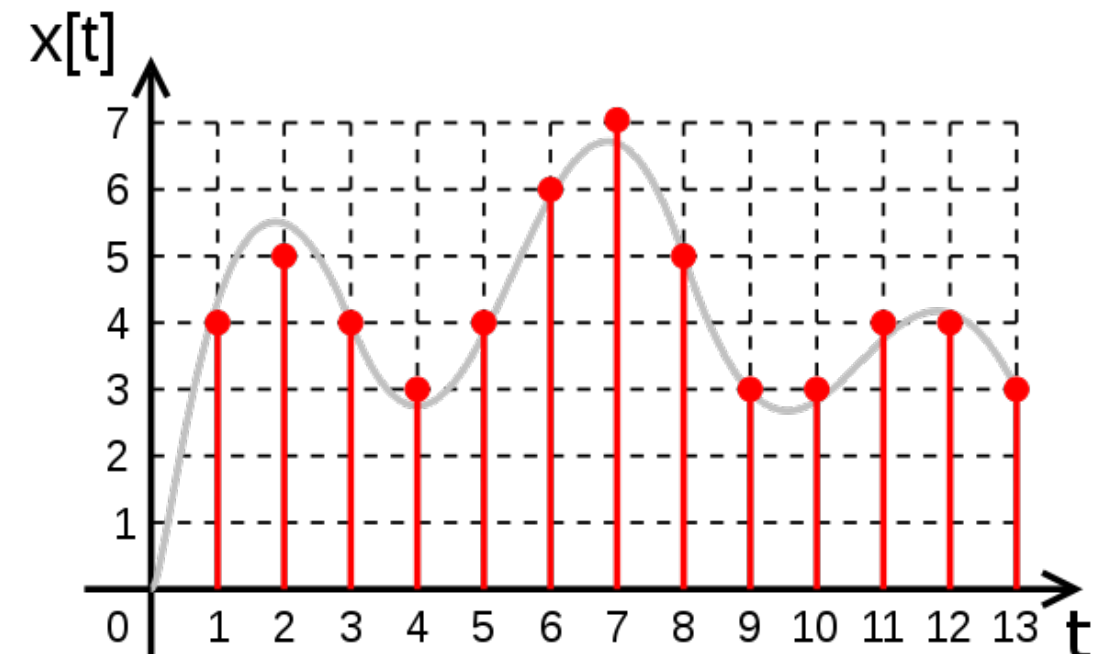
1991 Mathematics Subject Classification: 52C17

Keywords and Phrases: Sphere packings; lattices; quadratic forms; geometry of numbers



1 INTRODUCTION

The sphere packing problem has its roots in geometry and number theory (it is part of Hilbert's 18th problem), but is also a fundamental question in information theory. The connection is via the sampling theorem. As Shannon observes in his classic 1948 paper [37] (which ushered in the age of digital communication), if f is a signal of bandwidth W hertz, with almost all its energy concentrated in an interval of T secs, then f is accurately represented by a vector of $2WT$ samples, which may be regarded as the coordinates of a single point in $\mathbb{R}^n$, $n = 2WT$. Nearly equal signals are represented by neighboring points, so to keep the signals distinct, Shannon represents them by n -dimensional 'billiard balls', and is therefore led to ask: what is the best way to pack 'billiard balls' in n dimensions?



In physics, hard spheres are the simplest model for liquids, glasses, colloids, granulars, ...

But: physics focuses on 2d and 3d, while information theory focuses on large d

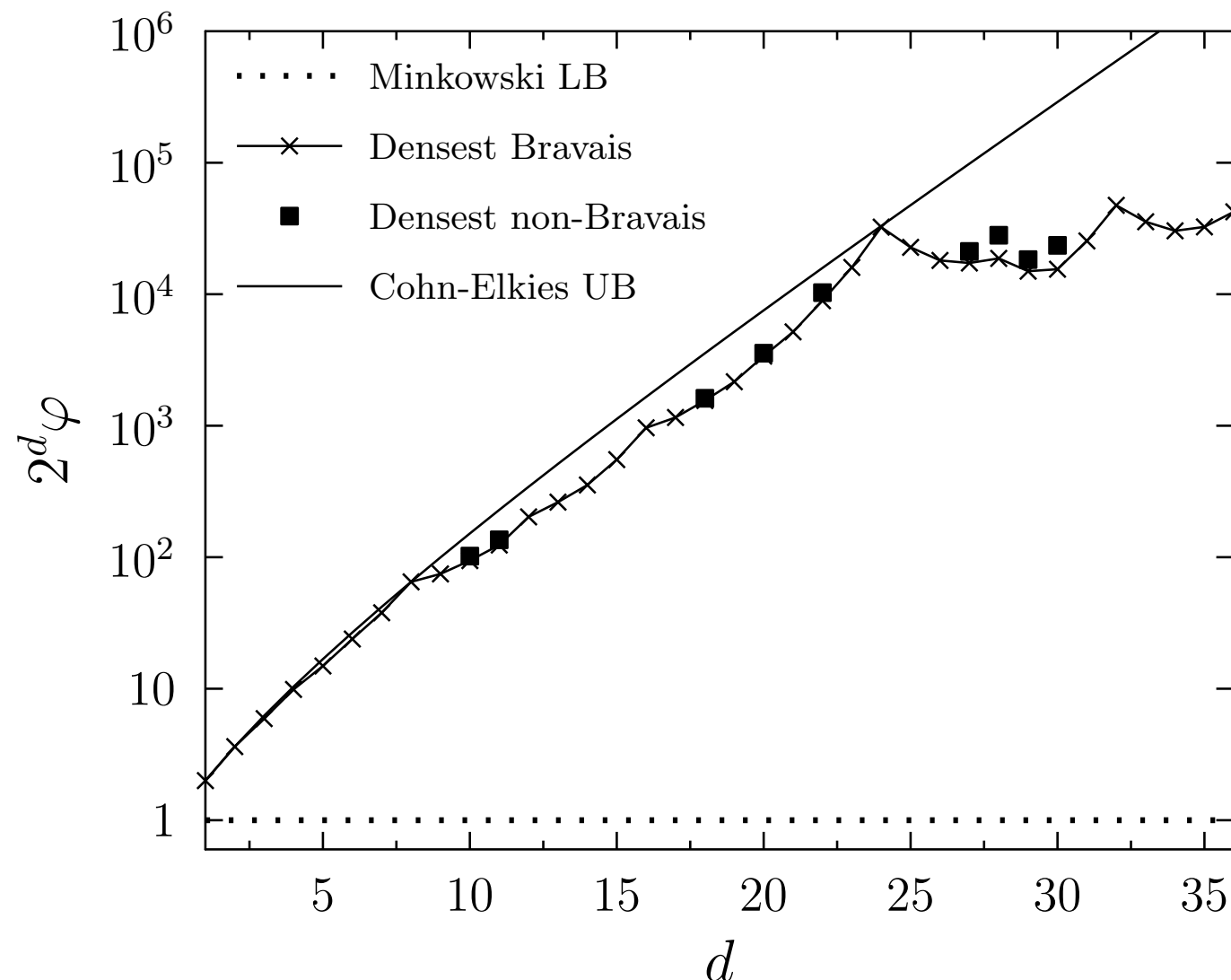
Approaches

Many possible approaches:

- Look directly for very dense packings. Typically Bravais lattice packings.
- Prove upper and lower bounds on best packing density
- Consider the uniform probability measure over all packings of given density (equilibrium hard spheres). Try to sample from it.
- Consider a probability measure over a space of lattices. Try to sample from it.
- Consider non-equilibrium search algorithms of various nature

Bounds and lattices

Known lattices



- **Minkowski 1905:** $\varphi \geq 2^{-d}$

[The proof is actually elementary. By definition, if we double the radius of each sphere, then space must be covered completely; otherwise, a new sphere would fit in the original packing. Doubling the radius multiplies volume by 2^d , and so the original packing must cover at least a 2^{-d} fraction of space.]

- **Kabatiansky-Levensthein:** $\varphi \leq 2^{-0.5990d}$
- **Cohn-Elkies 2003:** improved at finite d

- **Proven densest packings:**

$d = 2$ (Thue, 1892)

$d = 3$ (Hales, 1998-2005)

$d = 8$ (Viazovska, 2016)

$d = 24$ (Cohn et al. 2016)

Remark 1: the proofs are when the densest known packing touches the upper bound

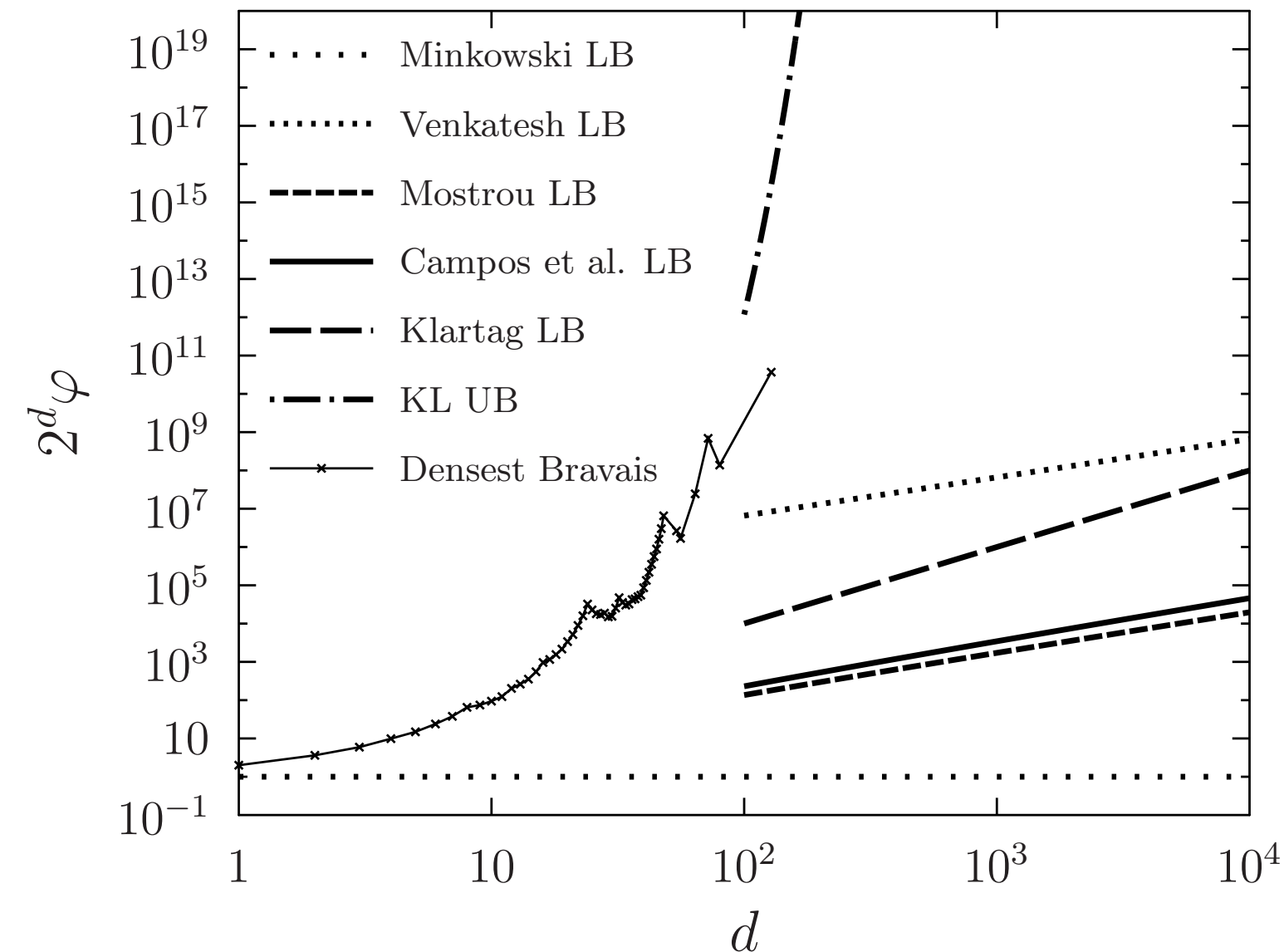
Remark 2: in some dimensions, the best known packing is non-Bravais

Remark 3: the behavior is far from being smooth due to symmetries

Nebe and Sloane, 2015. Table of densest packings presently known

www.math.rwth-aachen.de/~Gabriele.Nebe/LATTICES/density.html

Lower bounds



- **Minkowski 1905:** $\varphi \geq 2^{-d}$

[The proof is actually elementary. By definition, if we double the radius of each sphere, then space must be covered completely; otherwise, a new sphere would fit in the original packing. Doubling the radius multiplies volume by 2^d , and so the original packing must cover at least a 2^{-d} fraction of space.]

- **Rogers 1947:** $\varphi \geq c d 2^{-d}$

[Constructive: sample random Bravais lattices. The value of c improved from $c = 2/e$ (Rogers 1947) to $c = 6/e$ (Vance 2011) and $c = 65,963$ (Venkatesh 2012).]

- **Venkatesh 2012:** $\varphi \geq c \log(\log d) d 2^{-d}$

[With $c = 1/2$. Moustrou (2017) updated to $c = 0.89$ and presented a deterministic construction with $\exp(1.5d \log d)$ binary operations.]

- **Campos et al. 2023:** $\varphi \geq 0.5 \log d d 2^{-d}$

[Constructive and disordered: construct random graph and its maximum independent set.]

- **Klartag 2025:** $\varphi \geq c d^2 2^{-d}$

[Constructive: greedy algorithm for Bravais lattices. Note the constant c is not bounded in the proof, I used $c = 1$ in the plot.]

Remark 1: best LB and UB are exponentially divergent

Remark 2: known packings seem to be somewhere in between

Nebe and Sloane, 2015. Table of densest packings presently known

www.math.rwth-aachen.de/~Gabriele.Nebe/LATTICES/density.html

Random Bravais lattices

Random Bravais lattices

$$\vec{x}_{\vec{n}} = n_1 \vec{a}_1 + \cdots n_d \vec{a}_d = A \vec{n} \quad \vec{n} \in \mathbb{Z}^d$$

$$A = (\vec{a}_1, \cdots, \vec{a}_d)$$

Volume of the unit cell is $|\det A|$

Closest point to the origin is at distance $D = \min_{\vec{n}} |A \vec{n}|$

Packing of spheres of radius $R = D/2$

Sphere $\mathcal{B}_d$ of radius D : $\mathcal{B}_d = \{\vec{x} : |\vec{x}| \leq D\}$

Packing fraction $2^d \varphi = \text{vol}(\mathcal{B}_d) / |\det A|$

Equivalently $\vec{y} = A^{-1} \vec{x}$ with cubic lattice $\vec{n}$

Ellipsoid $\mathcal{E}_d = A^{-1} \mathcal{B}_d = \{\vec{y} : |A \vec{y}|^2 \leq 1\}$

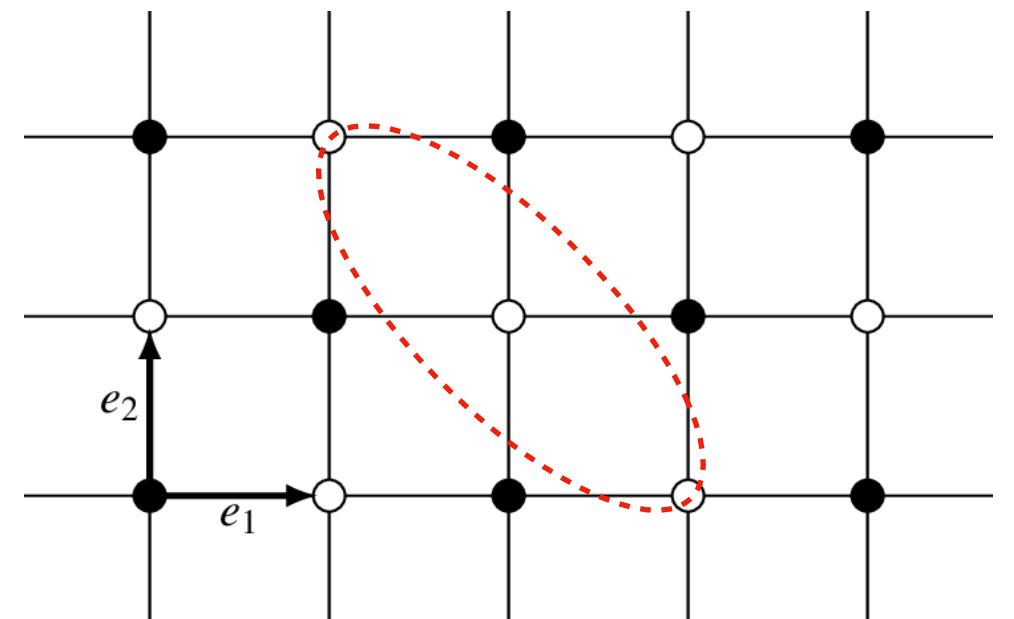
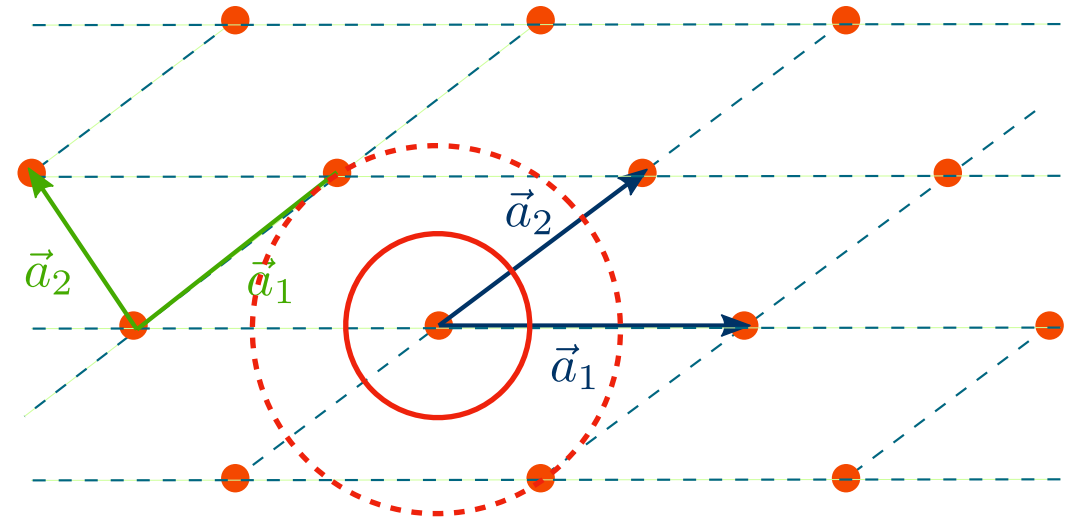
Problem: maximize $2^d \varphi = \text{vol}(\mathcal{E}_d)$

Note: $\mathcal{E}_d = \{\vec{y} : \vec{y} G \vec{y} \leq 1\}$

The symmetric matrix $G = A^T A$ has $d(d+1)/2$ parameters:
 d axis dilation and $d(d-1)/2$ rotations

Because of $\vec{y} \leftrightarrow -\vec{y}$ symmetry, boundary points come in pairs

Expect $d(d+1)$ points on the boundary



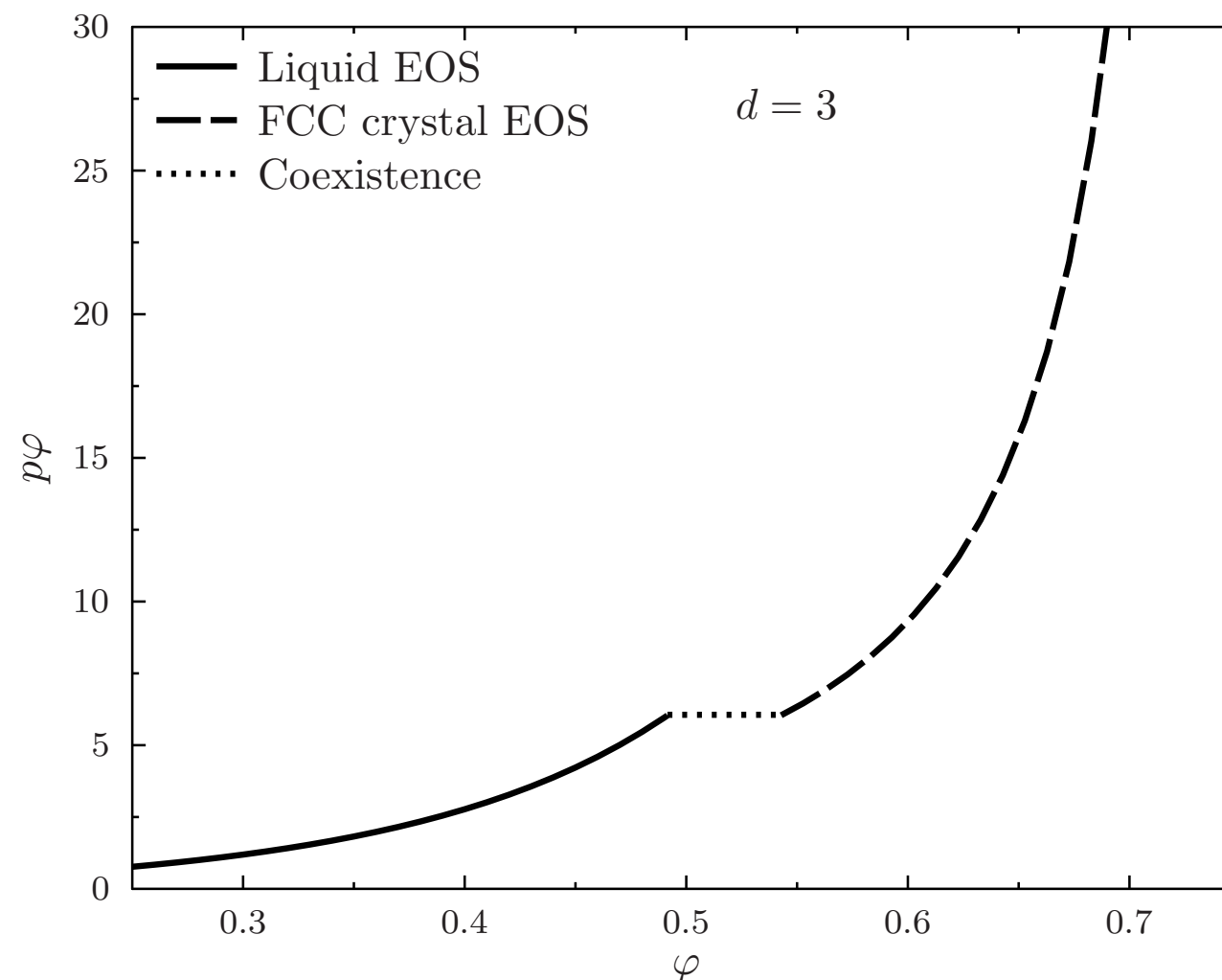
Random Bravais lattices - results

- Rogers (1947) - 'uniform' measure over lattices
Rigorously proved that with positive probability $\text{vol}(\mathcal{E}_d) \geq cd$
- Parisi (2008) - attempt to compute $\langle e^{\beta \text{vol}(\mathcal{E}_d)} \rangle_A$ over the Rogers measure - no better results
- Kallus-Elser-Gravel (2010) and Andreanov-Scardicchio (2012) attempted numerical sampling of lattices. Found known packings up to $d = 19$
- Marcotte-Torquato (2013) and Kallus-Marcotte-Torquato (2013) devised a greedy algorithm to maximize $\text{vol}(\mathcal{E}_d)$
Found known best packings up to $d = 24$. Fitting suggests $\text{vol}(\mathcal{E}_d) \sim cd^3$
- Kallus (2013) numerically sampled from $e^{\beta \text{vol}(\mathcal{E}_d)}$. Found known best packings up to $d = 20$ by simulated annealing. Found evidence for a first-order transition along the annealing.
- Klartag (2025) rigorously proved that a greedy algorithm can achieve $\text{vol}(\mathcal{E}_d) \geq cd^2$

Remark: finding $D = \min_{\vec{n}} |A\vec{n}|^2$ is an NP-complete problem (Ajtai 1997)

Hard spheres in equilibrium

The 3d phase diagram



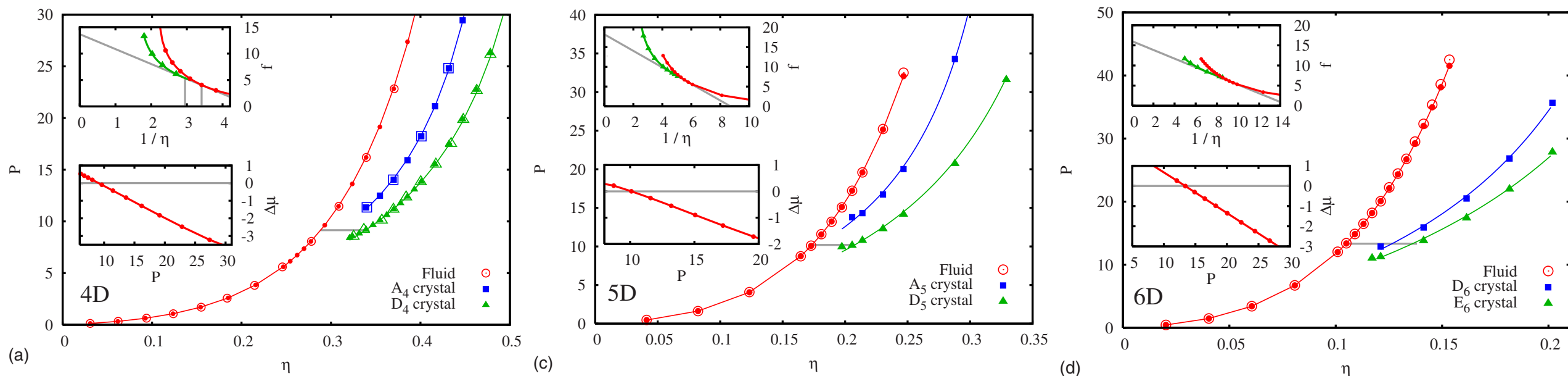
Low density: liquid phase, proven to exist via cluster expansion (Lebowitz-Penrose 1964)

High density: FCC/BCC at 74% packing fraction, rigorously proven (Hales 1998-2005)

Non-rigorous, phenomenological equations of state for both liquid and crystal

First order transition liquid-crystal well established numerically, very far from any proof

The 4d, 5d, 6d... phase diagram



Van Meel, Charbonneau, Fortini, Charbonneau (2009)

It becomes more and more difficult to even simulate the liquid phase:
generically scales as $d^{d/2}$, practically limited to $d \leq 13$.

Known crystals can be prepared at close packing and simulated to lower pressure.
First order transition.

Nucleation of the crystal becomes more and more difficult.
Already in $d = 4$ the liquid phase is strongly metastable.
No spontaneous nucleation can be observed in $d > 4$.

Remark 1: Obtaining crystals by slow compression seems a really bad idea

Remark 2: Crystals with lower close packing have higher pressure and lower entropy

The 4d, 5d, 6d... 10d phase diagram

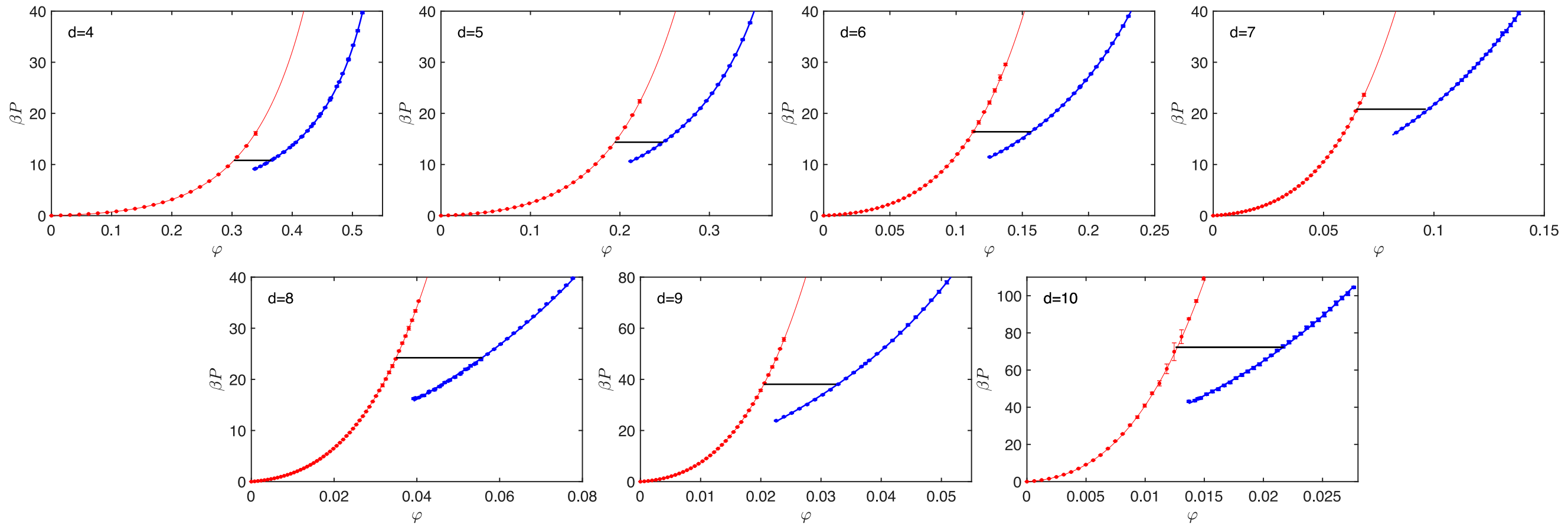


Figure 2. Liquid (red) and crystal (blue) equations of state and coexistence conditions (black) in $d = 4-10$. Pressures from numerical simulations (points) of the liquid are obtained for $N = 2048, 3888, 3000, 3000, 1000, 9000$, and 6000 in $d = 4-10$ respectively (where $d = 8-10$ are embedded in E_8 , D_9 , and D_{10} respectively) and are in quantitative agreement with the [4,5] Padé approximant (line) in $d = 4-9$ and the [5,6] Padé approximant in $d = 10$ of the virial expansion (Eq. (5)) [14–17]. Pressures from numerical simulations (points) of the crystal are obtained for 2048, 3888, 5184, 17496, 6561, 39366, and 81920 in $d = 4-10$ respectively. Fits to the empirical form given in Eq. (6) (solid blue line) quantitatively recapitulate the simulation results. Coexistence conditions are such that $\mu_\ell = \mu_s$ and $P_\ell = P_s$, as described in Sec. VI. Error bars denote 95% confidence intervals. Pressure is reported in units that implicitly set the particle diameter to unity, i.e., with $\sigma = 1$.

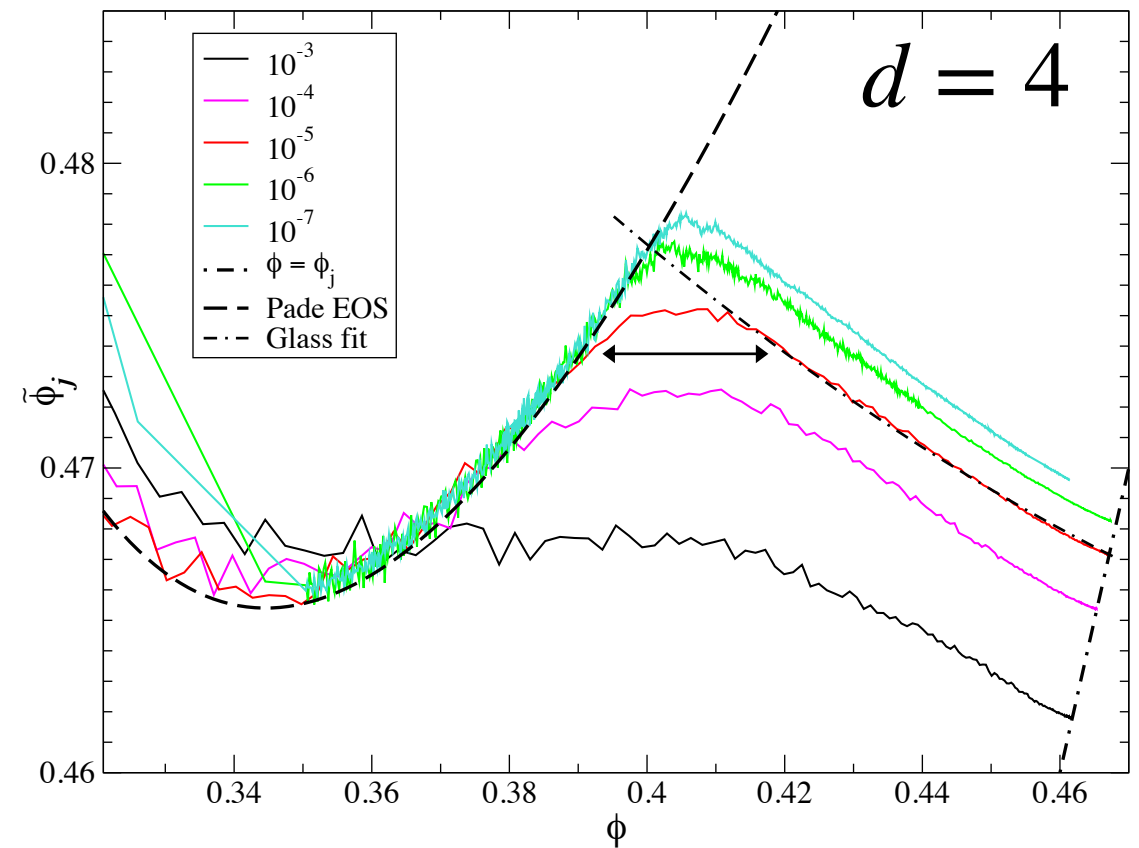
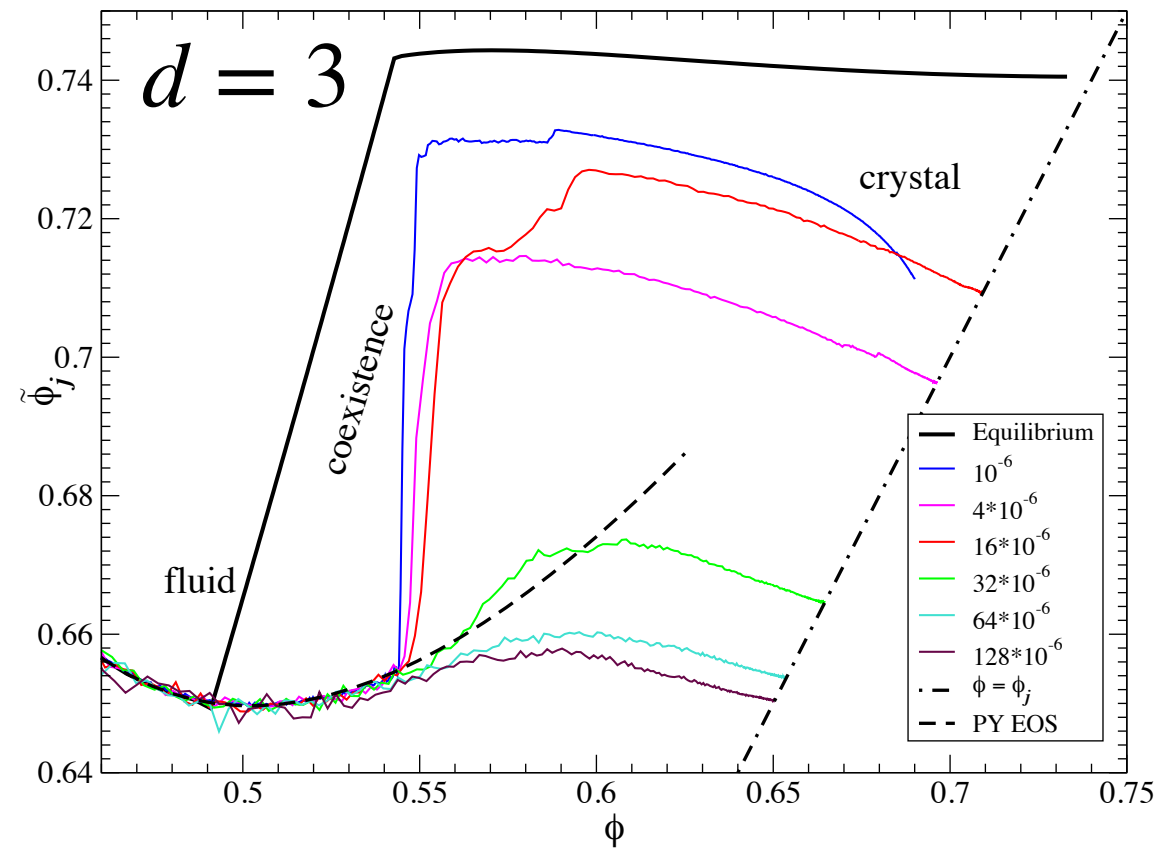
Charbonneau, Gish, Hoy, Morse (2021)

Remark: No guarantee that these are the densest crystals except in $d=3,8$ (and 4)



Hard spheres glass

The $d \rightarrow \infty$ liquid/glass branch



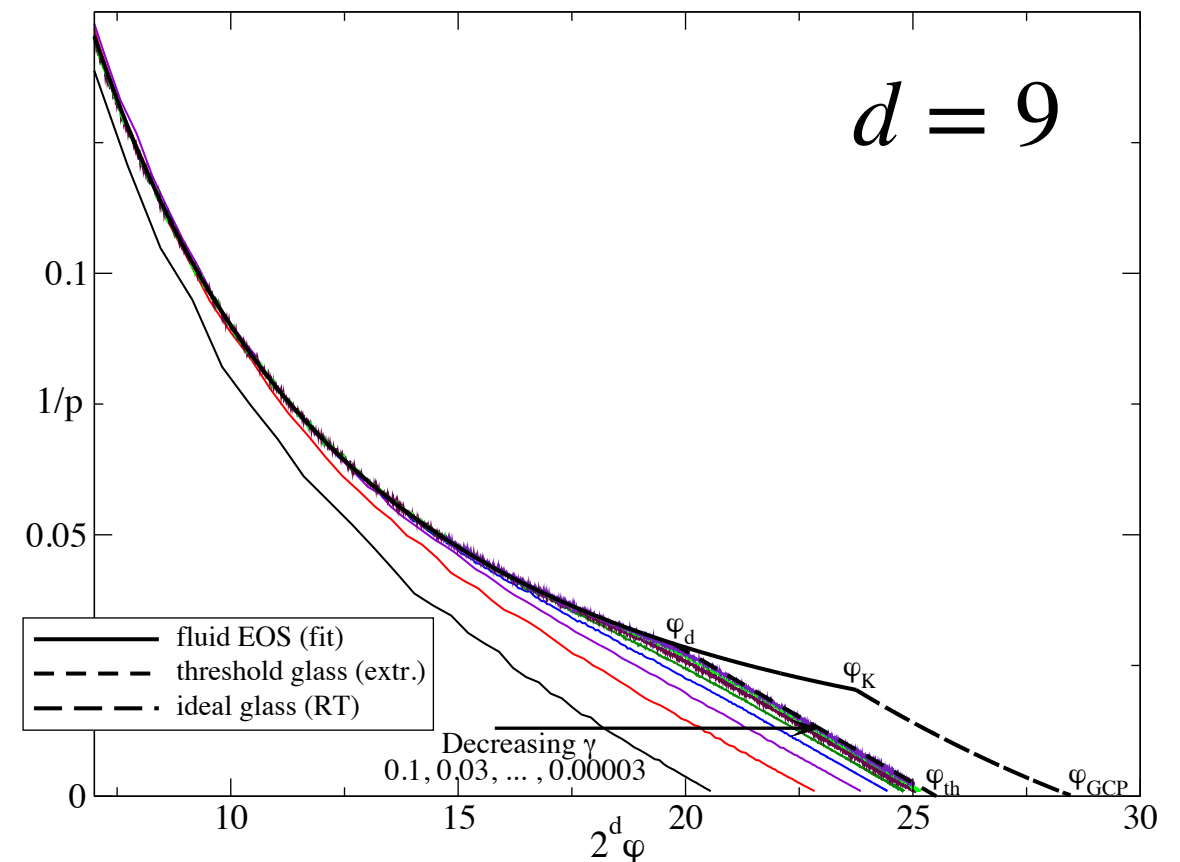
Skoge, Donev, Stillinger, Torquato (2006)

Metastable liquid branch
becomes reproducible

Glass transition well visible

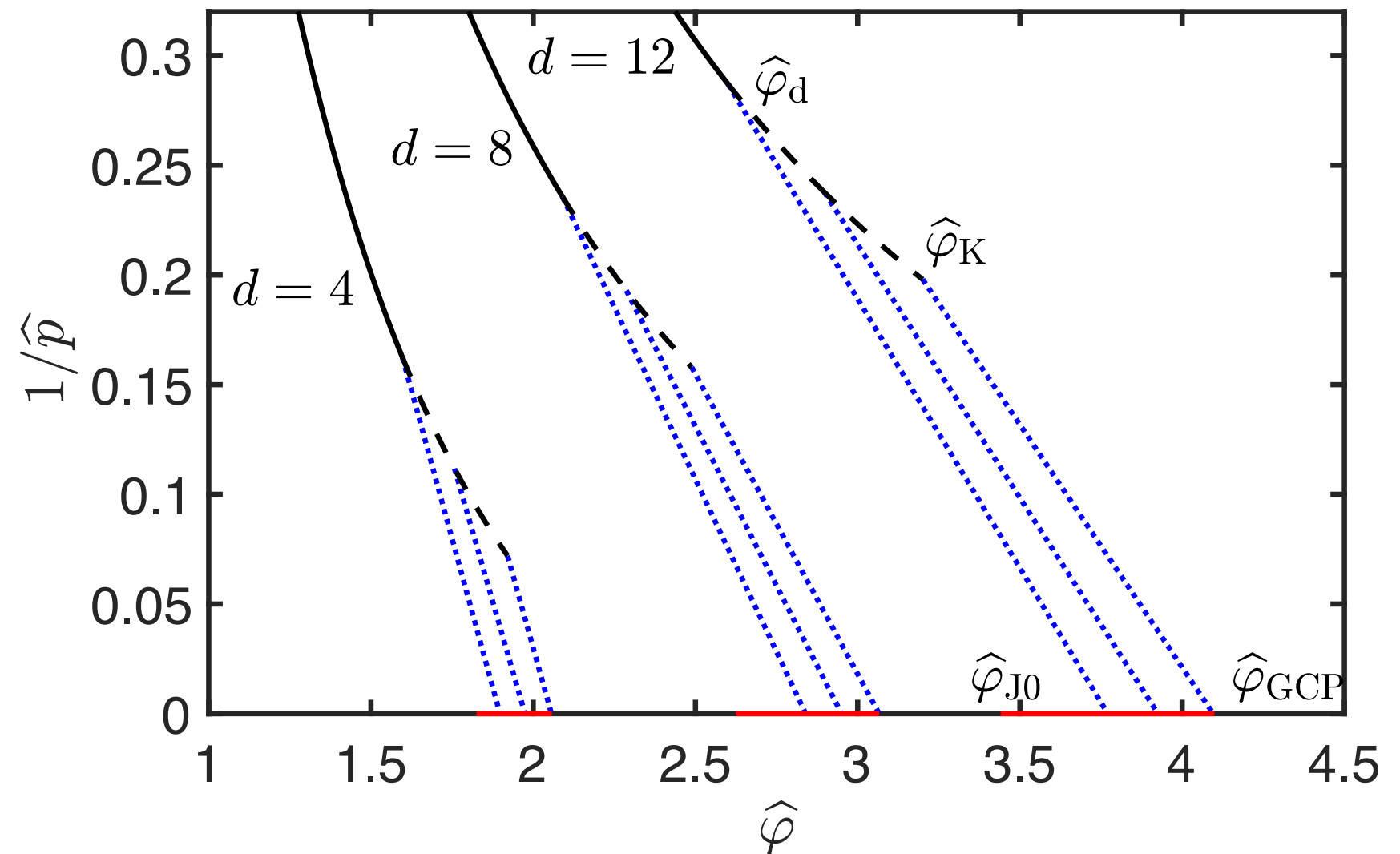
Theory: ideal glass transition
and glass close packing via
replica method

Parisi, Zamponi (2010)
Mangeat, Zamponi (2016)



Charbonneau, Ikeda, Parisi, Zamponi (2011)

The $d \rightarrow \infty$ liquid/glass branch



Metastable liquid branch
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Parisi, Zamponi (2010)
Mangeat, Zamponi (2016)

Charbonneau, Hu, Kundu, Morse (2022)

Charbonneau, Morse (2021)

Liquid/glass branch terminates at $\varphi_{GCP} \sim \log d d 2^{-d}$

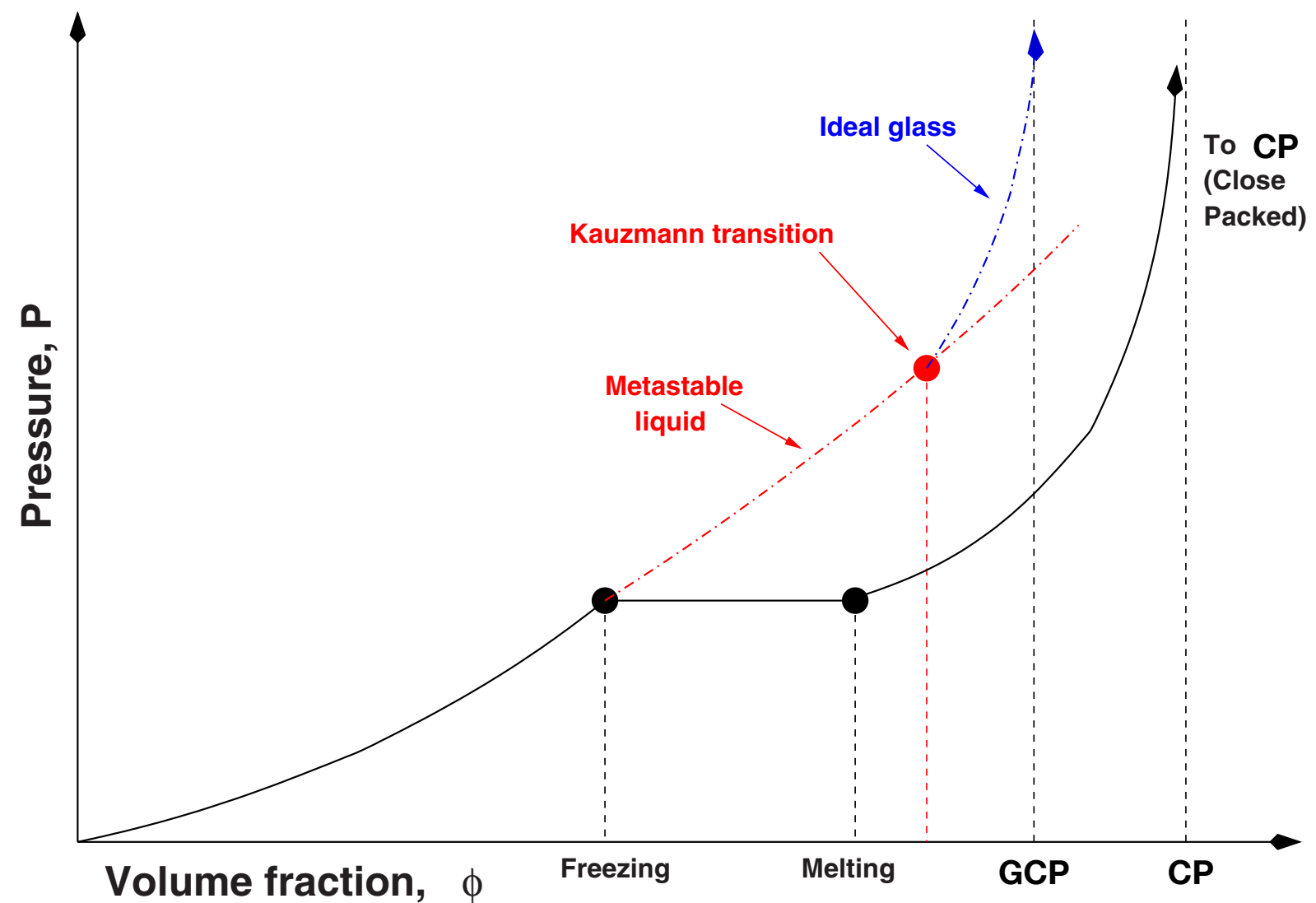
Non-trivial critical exponents around jamming predicted
by the theory, in all dimensions

Charbonneau, Kurchan, Parisi, Urbani, Zamponi (2014)



Summary

The $d \rightarrow \infty$ phase diagram



Disordered branch: liquid, then glass at $\phi_d \sim 4.8d2^{-d}$, then GCP at $\phi_{\text{GCP}} \sim \log d d2^{-d}$

Local search algorithms starting from low-density liquid are not supposed to beat GCP

Parisi, Zamponi (2010)

Very recently found crystal branch with $\phi_{\text{CP}} \sim cd^22^{-d}$ (Klartag 2025)

Implies $\phi_f \sim \log d d2^{-d}$ and $\phi_m \sim (\log d)^3 d2^{-d-1}$

Freezing would happen in the glass phase, nucleation is impossible

Charbonneau, Morse, Perkins, Zamponi (2021)

The $d \rightarrow \infty$ phase diagram

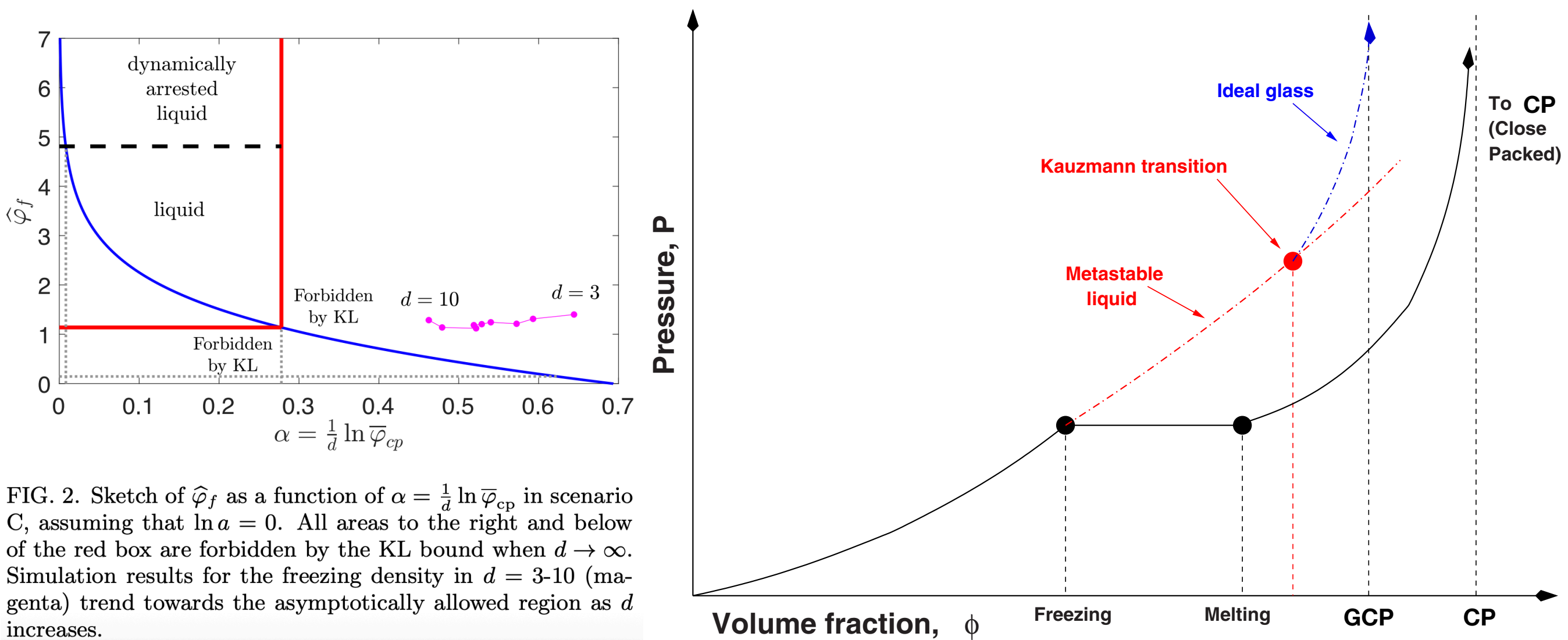


FIG. 2. Sketch of $\hat{\varphi}_f$ as a function of $\alpha = \frac{1}{d} \ln \bar{\varphi}_{cp}$ in scenario C, assuming that $\ln a = 0$. All areas to the right and below of the red box are forbidden by the KL bound when $d \rightarrow \infty$. Simulation results for the freezing density in $d = 3-10$ (magenta) trend towards the asymptotically allowed region as d increases.

Higher-density crystals are possible

Torquato, Stillinger (2010)

Crystal branch with $\varphi_{CP} \sim cd^{\psi+1} 2^{-d}$ still has $\varphi_f \sim \log d d 2^{-d}$

Crystal branch with $\varphi_{CP} \sim ce^{\alpha d} 2^{-d}$ would have $\varphi_f \sim Ad 2^{-d}$

Charbonneau, Morse, Perkins, Zamponi (2021)



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Thank you for your attention!

Make d -dimensional lattice with k lattices in $\tilde{d} = d/k$

Vectors $\vec{b}_1, \dots, \vec{b}_{\tilde{d}}$ in dimension $\tilde{d} = d/k$

Volume of the unit cell is $|\det A| = |\det B|^k$

Closest point to the origin is at distance $D = \min_{n_1, \dots, n_d} |A\vec{n}|^2 = \min_{n_1, \dots, n_{\tilde{d}}} |B\vec{n}|^2 = \tilde{D}$

$$\bar{\varphi} = 2^d \varphi = \frac{V_d D^d}{|\det A|} = \frac{V_d (\tilde{D}^{d/k})^k}{|\det B|^k} \quad \text{and} \quad \widetilde{\varphi} = 2^{\tilde{d}} \tilde{\varphi} = \frac{V_{d/k} \tilde{D}^{d/k}}{|\det B|}$$

$$\text{hence using } V_d = \frac{\pi^{d/2}}{\Gamma(d/2 + 1)} \quad \text{we have} \quad \bar{\varphi} = \frac{V_d}{(V_{d/k})^k} (\widetilde{\varphi})^k = \frac{\Gamma[d/(2k) + 1]^k}{\Gamma(d/2 + 1)} (\widetilde{\varphi})^k$$

If $d/k = \tilde{d}$ fixed and $d \rightarrow \infty$ then $\widetilde{\varphi}$ is constant and

$$\bar{\varphi} = V_d \times (\widetilde{\varphi} / V_{\tilde{d}})^{d/\tilde{d}}$$

Then V_d goes to zero as a factorial while the other term is exponential in d

$$\text{If } k \text{ fixed then } \bar{\varphi} = k^{-d/2} (d)^{k/2-1/2} \times k^{k/2} (\pi)^{k/2-1/2} (\widetilde{\varphi})^k$$

$$\text{If } \widetilde{\varphi} \sim \tilde{d}^\alpha \sim d^\alpha \text{ then } \bar{\varphi} \rightarrow 0$$

$$\text{If } \widetilde{\varphi} \sim e^{a\tilde{d}} \sim e^{ad/k} \text{ then } \bar{\varphi} \sim e^{d[a-(\log k)/2]} \text{ has a worse growth}$$

Klartag's proof

Ellipsoid $\mathcal{E}_t = \{\vec{y} : |A_t \vec{y}|^2 \leq 1\} = \{\vec{y} : \vec{y} G_t \vec{y} \leq 1\}$

$$G_t = A_t^T A_t \in \mathbb{R}_+^{d \times d}$$

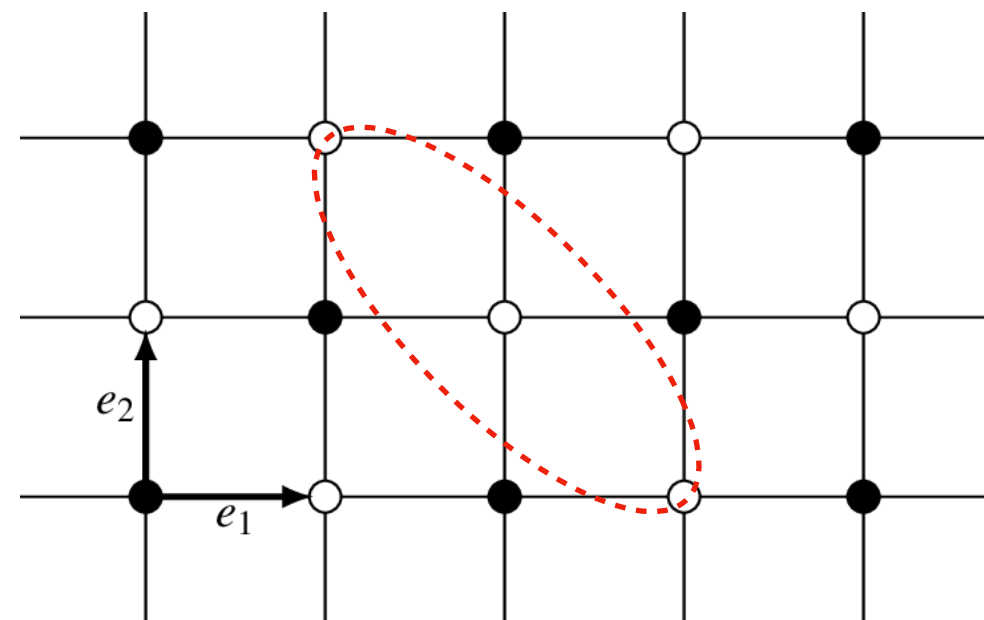
Problem: maximize $\text{vol}(\mathcal{E}_t) = V_d / \sqrt{\det G_t}$

Euclidean space $\mathbb{R}_{\text{symm}}^{d \times d}$ with $A \cdot B = \text{Tr}[AB]$

W_t Dyson Brownian motion

$$F_t = \{B \in \mathbb{R}_{\text{symm}}^{d \times d} : \vec{x} B \vec{x} = 0, \forall \vec{x} \in L : \vec{x} G_t \vec{x} = 1\}$$

π_t projector on F_t and $\tilde{\pi}_t = \mathbb{I} - \pi_t$ and $N_t = \dim(F_t) = \text{Tr}(\pi_t)$



Algorithm:

1. Initialize $G_0 = g_0 \mathbb{I}$ with g_0 small enough to be L-free
2. Do $G_t = G_0 + W_t$ until touching a first lattice point $\vec{x}_1 G_{\tau_1} \vec{x}_1 = 1$ (and $-\vec{x}_1$) at time τ_1
3. Do $G_t = G_{\tau_1} + \pi_{\tau_1}(W_t)$ until touching a second lattice point $\vec{x}_2 G_{\tau_2} \vec{x}_2 = 1$ at time τ_2
4. Iterate: do $G_t = G_{\tau_{i-1}} + \pi_{\tau_{i-1}}(W_t)$ until touching a new lattice point $\vec{x}_i G_{\tau_i} \vec{x}_i = 1$ at time τ_i
5. Stop when $F_t = \emptyset$

Technical lemmas. Almost surely: (1) G_t is continuous, positive-definite and L-free; (2) one point is added at each step; (3) the times $\{\tau_i\}$ are finite; (4) the number of steps is finite; (5) at the end there are $d(d+1)/2$ pairs $\pm \vec{x}$ on the boundary

Main ingredients:

- (1) The Brownian motion almost surely drives one eigenvalue to zero in a finite time
- (2) A continuous process G_t not always positive must gain a contact point in a finite time

Klartag's proof

Ellipsoid $\mathcal{E}_t = \{\vec{y} : |A_t \vec{y}|^2 \leq 1\} = \{\vec{y} : \vec{y} G_t \vec{y} \leq 1\}$

$$G_t = A_t^T A_t \in \mathbb{R}_+^{d \times d}$$

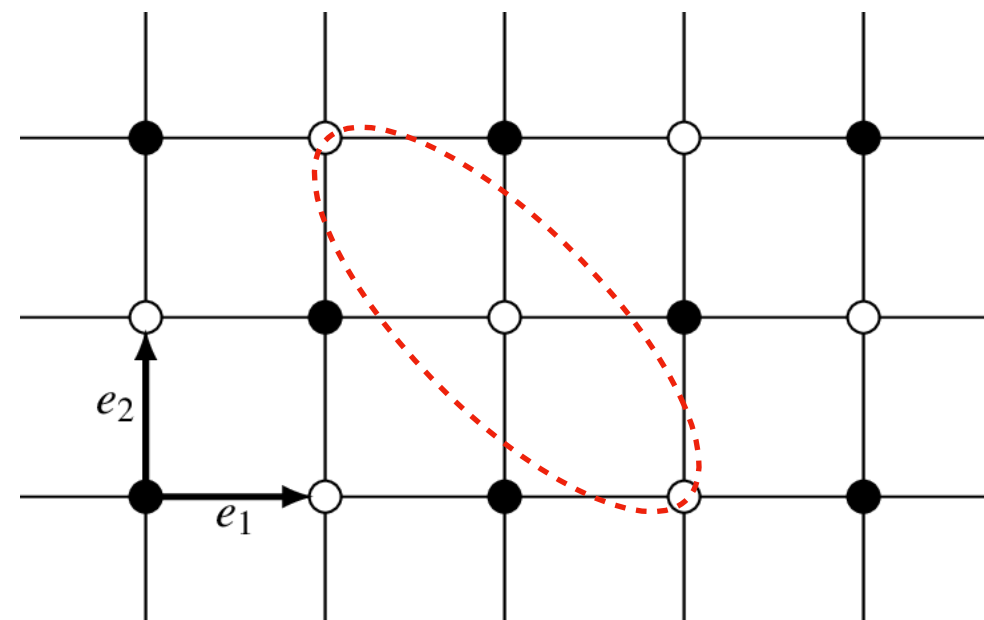
Problem: maximize $\text{vol}(\mathcal{E}_t) = V_d / \sqrt{\det G_t}$

Euclidean space $\mathbb{R}_{\text{symm}}^{d \times d}$ with $A \cdot B = \text{Tr}[AB]$

W_t **Dyson Brownian motion**

$$F_t = \{B \in \mathbb{R}_{\text{symm}}^{d \times d} : \vec{x} B \vec{x} = 0, \forall \vec{x} \in L : \vec{x} G_t \vec{x} = 1\}$$

π_t **projector on** F_t and $\tilde{\pi}_t = \mathbb{I} - \pi_t$ and $N_t = \dim(F_t) = \text{Tr}(\pi_t)$



$$dG_t = \pi_t(dW_t) \quad \text{hence} \quad G_t = g_0 \mathbb{I} + \int_0^t \pi_s(dW_s) = g_0 \mathbb{I} + \frac{W_t^+ + W_t^-}{2}$$

$$W_t^\pm = \int_0^t \pi_s(dW_s) \pm \int_0^t \tilde{\pi}_s(dW_s) \text{ are two Dyson Brownian motions}$$

Norm $\|G_t\| = \sup |G \vec{x}| / |\vec{x}|$ and then $P(\|G_t - g_0 \mathbb{I}\| \geq C_0 r) \leq C \exp(-r^2/t)$

$$d(\log \det G_t) = \text{Tr}[G_t \pi_t(dW_t)] - \frac{1}{2} \delta_t dt \quad \text{with} \quad \delta_t \geq \frac{1}{\|G_t\|^2} \text{Tr}[\pi_t]$$

$$\mathbb{E}[\log \det G_t] \leq d \log g_0 - \frac{1}{2} \int_0^t \mathbb{E}[\|G_s\|^{-2} N_s] ds$$